Assignment #3

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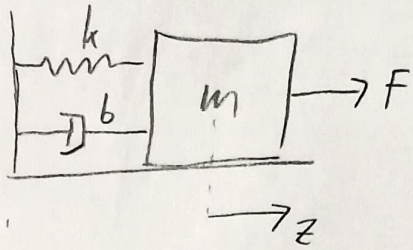
Problem D.3:

Find:
- PE of system, P
- Generalized coordinates, $\vec{q}$
- Generalized forces, $\vec{\tau}$
- Damping forces, $\vec{b}$
- EoM using Euler-Lagrange
- Simulation using variable force input



Solution:

By inspection, taking $P(z=0)=0$ to be unstretched spring...

$$P = \frac{1}{2}kz^2 \longleftarrow P$$

$$\vec{q} = [z] \longleftarrow \vec{q}$$

$$\tau = [F] \longleftarrow \vec{\tau}$$

$$\vec{b} = [-b\dot{z}] \longleftarrow \vec{b}$$

And since from the previous homework we have $K = \frac{1}{2}m\dot{z}^2$, then by Euler Lagrange, where $L = K - P = \frac{1}{2}m\dot{z}^2 - \frac{1}{2}kz^2$...

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\vec{q}}}\right) - \frac{\partial L}{\partial \vec{q}} = \vec{\tau} - B\dot{\vec{q}}$$

$$\Rightarrow \frac{d}{dt}([m\dot{z}]) - ([-kz]) = [F] + [-b\dot{z}]$$

$$\Rightarrow m\ddot{z} + kz = F - b\dot{z} \quad \Rightarrow \quad \ddot{z} = \frac{1}{m}[F - b\dot{z} - kz] \longleftarrow \text{EoM}$$

In state-space...

$$x = \begin{bmatrix} z \\ \dot{z} \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} \dot{z} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} \dot{z} \\ \frac{1}{m}(F - b\dot{z} - kz) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} x + \begin{bmatrix} 0 \\ 1/m \end{bmatrix}[F]$$

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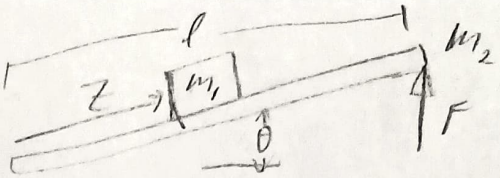
Problem E.3

Find: same as D.3



Solution:

By inspection...

$$\vec{q} = \begin{bmatrix} z \\ \theta \end{bmatrix}, \quad \dot{\vec{q}} = \begin{bmatrix} \dot{z} \\ \dot{\theta} \end{bmatrix} \quad \leftarrow \vec{q}$$

$$P = m_2 g\left(\frac{\ell}{2}\sin\theta\right) + m_1 g z \sin\theta \quad \leftarrow P$$

$$\vec{\tau} = \begin{bmatrix} 0 \\ F\ell\cos\theta \end{bmatrix} \quad \leftarrow \vec{\tau}$$

$$\vec{b} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \leftarrow \vec{b}$$

And using sympy and KE from HW 2...

$$\left.\begin{aligned} \ddot{z} &= -g\sin\theta + z\dot{\theta}^2 \\ \ddot{\theta} &= \frac{3\left(2F\ell\cos\theta - m_2 g \ell\cos\theta - 2m_1 g z\cos\theta - 4m_1 z\dot{\theta}\dot{z}\right)}{2\left(m_2\ell^2 + 3m_1 z^2\right)} \end{aligned}\right\} \leftarrow \text{EoM}$$

Problem F.3

Find: Same as D.3

Solution:

By inspection

$$\vec{q} = \begin{bmatrix} z_r \\ h \\ \theta \end{bmatrix} \quad \leftarrow \vec{q}$$

$$P = (m_c + m_\ell + m_r) g h \quad \leftarrow P$$

$$\vec{\tau} = \begin{bmatrix} -(f_r + f_\ell)\sin\theta \\ (f_r + f_\ell)\cos\theta \\ d(f_r - f_\ell) \end{bmatrix} \quad \leftarrow \vec{\tau}$$

$$\vec{b} = \begin{bmatrix} -\mu\dot{z}_r \\ 0 \\ 0 \end{bmatrix} \quad \leftarrow \vec{b}$$

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Using symmetry and KE from HW2...

$$\ddot{z}_v = \frac{-1}{m_c + 2m_r}\left(f_l \sin\theta + f_r \sin\theta + \mu \dot{z}_v\right)$$

$$\ddot{h} = \frac{1}{m_c + 2m_r}\left((f_l + f_r)\cos\theta - g(m_c + m_l + m_r)\right)$$

$$\ddot{\theta} = \frac{d(f_r - f_l)}{J_c + 2d^2 m_r}$$

EoM